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MILITARY SPECIFICATION

TRANSMITTER, RATE OF FLOW, (FUEL TRU-98/A)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for variable turbo-jet engine, fuel, rate of flow transmitters.

* 1.2 Classification. Transmitters shall be of the following types, as specified (see 6.2):

Type	Flow Range (PPH)
TRU-98/A-1	0- 8,000
TRU-98/A-2	0-10,000
TRU-98/A-3	0-12,000
TRU-98/A-4	0-15,000 (second harmonic)
TRU-98/A-5	0-20,000
TRU-98/A-6	0-25,000
TRU-98/A-7	0-15,000 (second harmonic)
TRU-98/A-8	0-15,000 (second harmonic)

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards and handbooks. The following documents of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specifiction to the extent specified herein.

SPECIFICATIONS

Federal

PPP-B-636 Box, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/MMEDO, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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Military

MIL-P-116	Preservation, Methods of
MIL-C-5015	Connector, Electrical, Circular Threaded, AN Type, General Specification for
MIL-C-5541	Chemical Conversion Coatings on Aluminum: and Aluminum Alloys
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-C-7024	Calibrating Fluid, Aircraft Fuel System Components
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-S-20708	Synchros, General Specification for
MIL-C-26482	Connector, Electric, (Circular, Miniature, Quick Disconnect, Environment Resisting) Receptacles and Plugs, General Specification for
MIL-B-27497	Bearing, Jewel, Sapphire or Ruby, Synthetic
MIL-M-45202	Magnesium Alloy, Anodic Treatment of
MIL-I-83026	Indicator, Rate of Flow, Fuel EFU-26/A
MIL-C-83723	Connector, Electrical, (Circular, Environment Resisting), Receptacles and Plugs, General Specification for

STANDARDS

Military

DOD-STD-100	Engineering Drawing Practices
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-143	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-461	Electromagnetic Emission and Susceptability Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704	Aircraft Electric Power Characteristics
MIL-STD-781	Reliability Tests Exponential Distribution
MIL-STD-794	Parts and Equipment, Procedures for Packaging and Packing of
MIL-STD-810	Environmental Test Methods
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MS3106	Connector, Plug, Electric, Straight
MS3113	Connector, Receptacle, Electric, Series 1, Solder Type, Solder Mounting, Bayonet Coupling
MS3126	Connector, Plug, Electric, Crimp-Type, Straight, Bayonet Coupling
MS20756	Flange, Swivel, Retaining
MS33649	Bosses, Fluid Connection - Internalstraight Thread

(Copies of specifications, standards, handbooks, drawings and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

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3. REQUIREMENTS

* 3.1 First Article. When specified, a sample shall be subjected to first article inspection (see 4.4 and 6.6).

3.2 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143.

3.3 Materials.

3.3.1 Nonmagnetic materials. Nonmagnetic materials shall be used for all parts of the transmitter except where magnetic materials are essential.

3.3.2 Nonferrous materials. Nonferrous materials shall be used for all parts of the transmitter except where ferrous materials are essential.

3.3.3 Metals. Metals shall be of the corrosion-resistant type or suitably treated to resist corrosion due to fuels, salt spray, or atmospheric conditions likely to be met in storage or normal service.

3.3.4 Dissimilar metals. Unless suitably protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.3.4.1 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

3.3.5 Toxic and corrosive fumes. The transmitter shall be of such material and construction that it will not produce corrosive or deleterious fumes, acid, or alkali under normal use or under the conditions specified herein.

3.3.6 Protective treatment. When materials are used in the construction of the transmitter that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the requirements of this specification. The use of protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.3.7 Titanium. Where high temperature and other stringent environmental conditions are a factor in the design of the transmitter, the use of titanium will be permitted, with the approval of the procuring activity where lightweight and high strength materials are required. Use of lightweight materials and weight-saving designs shall be of major consideration and their use shall be investigated and exploited to the greatest possible extent.

3.4 Design and construction. Unless otherwise specified, the requirements of this specification are applicable to all types of transmitters.

* 3.4.1 Fuel variation. The transmitters shall be designed to operate satisfactorily with fuels having any of the variations in characteristics specified in MIL-T-5624. The fuel flow transmitter shall be calibrated for

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rate of flow in pounds per hour. MIL-C-7024C Type II calibration fluid may be used for testing purposes. The calibration of the transmitter shall be such that it will satisfactorily meet the requirements specified herein when tested with fuels in accordance with MIL-T-5624. The transmitter shall measure the fuel flow rate on a gravimetric basis without separate and distinct density correcting means.

* 3.4.2 System compatibility. The TRU-98/A-1 through /A-8 transmitters shall be interchangeable with previously manufactured TRU-98/A-1 through /A-8 transmitters of the same type and shall be completely compatible with the EFU-26/A-1 through /A-13 indicators conforming to MIL-I-83026 for the corresponding flow ranges.

3.4.3 Power. The transmitter shall be designed to operate from 115V ± 10 percent 400 Hz ± 10 percent single phase AC power having characteristics in accordance with MIL-STD-704. The transmitters shall meet the performance criteria specified herein when the input power is varied at combinations of voltages and frequencies ranging from 104 to 126V and 320 to 480 Hz and voltage spikes appearing up to 195V. The transmitter shall not be damaged by the application of voltages and frequencies of 10 percent more or less than the specified operating range. The TRU-98/A-7 transmitter shall be designed to operate over a frequency range of 400 Hz ± 20 percent.

3.4.4 Operation. The transmitter shall provide only one of the following types of output signals: A synchro output, a voltage output, or a second harmonic. It shall be capable of operating one or two compatible indicators. The signal impedance shall be such that suitable test equipment may be connected to the output signal leads for accurate fuel flow indication for engine trim purposes without disconnecting or disturbing the normally connected cockpit indicator.

* 3.4.4.1 Synchro output. Transmitters shall provide a five-wire synchro, conforming to MIL-S-20708, output for the indicators. The transmitter motor shall receive regulated power from the EFU-26/A indicator. The indicator power shall be one of the two types specified in MIL-I-83026. The transmitter and indicator shall operate as a system for the type of power used. The transmitter motor using the 8 Hz power shall be designed to operate from 8 Hz ± 0.05 percent 34 ± 4V rms 2 phase power, and the transmitter motor using the 400 Hz power shall be designed to operate from 400 ± 1 Hz 65 to 126V single phase power. The power shall be used to drive the transmitter impeller motor at a constant speed with the motor rotating the impeller rotor in a clockwise direction as viewed from the inlet end of the transmitter.

3.4.4.1.1 Electrical zero. Identification of the synchro leads and the determination of electrical zero shall be as follows:

a. The rotor leads shall be designated H_2 and C_2, and the C_2 lead shall be grounded.

b. One stator lead shall be designated Z_2 and connected to the C_2 lead.

c. Rated voltage shall be applied across H_2 and C_2, and the voltage measured across H_2 and one of the undesignated stator leads and across the two undesignated stator leads.

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d. The rotor shall be rotated until the voltage measured across H_2 and the one undesignated stator lead is approximately a minimum and the voltage across the two undesignated stator leads is exactly minimum. This position is the electrical zero position corresponding to one-half full scale indication.

e. The rotor shall be rotated in a clockwise direction from electrical zero as viewed from the rotor shaft end, the voltage across one of the undesignated stator leads and Z_2 shall increase, and the voltage across the other undesignated stator lead and Z_2 shall decrease. The stator lead for increasing voltage shall be labeled X_2 and the stator lead for decreasing voltage shall be labeled Y_2.

* 3.4.4.2 Voltage output. The transmitter shall be designed to provide a 400 Hz single phase voltage electrical output signal. The magnitude of the voltage shall be 5V for the full range of the transmitter. The electrical output signal shall have a carrier frequency identical to the input power frequency. The transmitter motor shall receive regulated power from the EFU-26/A indicator. The indicator power shall be one of the two types specified in MIL-I-83026. The transmitter and indicator shall operate as a system for the type of power used. The transmitter motor using the 8 Hz power shall be designed to operate from 8 Hz $\pm$ 0.05 percent 34 $\pm$ 4V rms 2 phase power, and the transmitter motor using the 400 Hz power shall be designed to operate from 400 $\pm$ 1 Hz, 65 to 126V single phase power. The power shall be used to drive the transmitter impeller motor at a constant speed with the motor rotating the impeller rotor in a clockwise direction as viewed from the inlet end of the transmitter.

* 3.4.4.3 Second harmonic output. The transmitter shall operate on the second harmonic principle in which the moving element consists of a nonenergized permanent 2 pole magnet (or magnets) and toroidally wound stator is energized with 115V 400 Hz single phase power. An externally located power supply shall provide 8 Hz $\pm$ 0.05 percent 34 $\pm$ 4V rms to drive the transmitter impeller motor at a constant speed with the motor rotating the impeller rotor in a clockwise direction as viewed from the inlet end of the transmitter.

3.4.4.4 Signal output versus rotation. For an increasing flow rate, the direction of rotation of the transmitter turbine rotor shall be clockwise as viewed from the inlet end of the transmitter. The pickoff rotor (turbine) shall rotate over two different linear ranges as defined in 3.5.6 with an expanded scale range having a rotation from 0° to 200° and a compressed scale range having a rotation from 200° to 307°, except for the TRU-98/A-7 transmitter which shall have an expanded scale range having a rotation from 0° to 204.7° and a compressed scale range having a rotation from 204.7° to 307°. The synchro transmitter shaft shall rotate in a clockwise direction, as viewed from the shaft end of the synchro and the inlet end of the transmitter, for an increasing flow rate.

3.4.4.5 Stop. A rugged mechanical stop shall be provided within the transmitter above the maximum flow rate value but below the zero point. The transmitter shall not be damaged by fuel-flow rates exceeding 20 percent of the rated value provided the pressure rating of the transmitter is not exceeded.

* 3.4.5 Operating pressure. The TRU-98/A-1 through /A-6 transmitters shall be designed for operation in fuel systems with a normal fuel pressure up to 1,000

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psi. It shall be capable of withstanding, without operational damage, a static fuel pressure of 1,600 psi at both room temperature and -54°C and a static fuel pressure of 1,000 psi at a fuel temperature of +71°C when in an ambient temperature of +140°C. The TRU-98/A-7 transmitter shall be designed to operate in fuel systems having a normal fuel pressure ranging from -9 to +60 psi and shall be capable of withstanding static fuel pressures ranging from -12 to +120 psi at +140°C ambient with a fuel temperature of +71°C. The TRU-98/A-8 transmitter shall be designed for operation in fuel systems with a normal fuel pressure of up to 800 psi. It shall be capable of withstanding, without operational damage, a static fuel pressure of 1360 psi at room temperature and -54°C, and 800 psi at +71°C with an ambient of +140°C.

* 3.4.6 Pressure drop. The TRU-98/A-1 through /A-6 transmitters shall be so designed that when inoperative under any fuel-flow conditions, the pressure drop will not be greater than when the transmitter is operating normally. Table I specifies the allowable pressure drop. The TRU-98/A-7 and /A-8 transmitters shall be designed for fuel flow up to 18,000 pph with pressure drops not exceeding those specified in table I.

3.4.7 Temperature range. The transmitter shall operate satisfactorily in ambient temperatures ranging from -54° to +121°C and high temperature exposure to 150°C. The operating fuel temperature range shall be from -54° to +71°C.

* 3.4.8 Electrical receptacles. The TRU-98/A-1 through /A-6 transmitters shall be provided with a hermetically sealed miniature electrical receptacle in accordance with MIL-C-5015 which shall be designed to meet the environmental requirements specified herein. The transmitters shall utilize a Cannon G-S02-16S-1P receptacle, or equivalent, for mating electrical plug MS3106E-16S-1S. The TRU-98/A-7 transmitter shall be provided with an electrical connector conforming to MIL-C-26482. The TRU-98/A-7 transmitter shall use a hermetically sealed connector, P/N M83723-88H-1407 per MIL-C-83723D, Class H.

* 3.4.9 Internal wiring. Internal wiring for all transmitters shall be identical. The wire shall be color coded and connected as indicated in table II. The system electrical connections shall be as shown on figure 1, 2, or 3 as applicable.

3.4.10 Reliability. The transmitter shall have a minimum acceptable mean-time-between-failures (MTBF) of 1,200 hours demonstrated to a 90 percent confidence level.

3.4.11 Longevity. The transmitter shall have an operating life span (equipment longevity as defined in MIL-STD-781) of not less than 2,500 hours before wearout failures occur or the transmitter consistently fails to meet the specified MTBF index. Parts requiring replacement within this period of time and the normal operating period of such parts shall be reported by the contractor to the procuring activity.

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TABLE I. Pressure drop.

Flow (PPH) TRU-98/A-1 thru /A-6	Pressure drop (PSI Max)
1,000	0.2
5,000	0.6
10,000	1.5
20,000	5.0
25,000	10.0
Flow (PPH) TRU-98/A-7 and /A-8	**Pressure drop (PSI Max)**
0 to 7,500	0.25
10,000	0.45
12,000	0.68
15,000	1.05
18,000	1.50

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*TABLE II. TRU-98/A Transmitter connections.

Synchro output			
	Pin	Function	Color Code
	A	C_2 lead of synchro rotor, low potential input of 115V 400 Hz regulated power to common, as applicable	Black/White
	B	H_2 synchro rotor lead (115V 400 Hz input)	Red/White
	C	X_2 synchro stator lead	Yellow
	D	Z_2 synchro stator lead	Black
	E	Y_2 synchro stator lead	Blue
400 Hz Power	F	115V 400 Hz single phase input from regulated power supply to motor lead	Black
400 Hz Power	G	Internal Ground	
	or		
8 Hz Power	F	34 ± 4V rms 8 Hz 2 phase input power to motor	White/Gray
8 Hz Power	G	34 ± 4V rms 8 Hz 2 phase input power to motor	White/Black
		Voltage output	
	Pin	Function	Color code
	A	Input power to pickoff, 115V 400 Hz ground	Green
	B	Input power to pickoff, 115V 400 Hz positive	Red
	C	Voltage output	Blue
	D	Voltage output	White
	E	8 Hz 34 ± 4V rms 2 phase input power to motor	White/Gray
8 Hz	F	8 Hz 34 ± 4V rms 2 phase input power to motor	White/Black
	G	Motor common	Black
	or		
	E	115V 400 Hz single phase input from regulated power supply to motor lead	Black

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TABLE II. TRU-98/A Transmitter connections (Cont).

Voltage output			
	Pin	Function	Color code
400 Hz	F	Internal ground	
	G	Spare	
		Second Harmonic Output (-4, -7 and -8 only)	
	Pin	Function	Color code
	A	Input power to pickoff, 115V 400 Hz ground	Green
		Input power to motor, ground	White/Yellow
	B	Input power to pickoff, 115V 400 Hz positive	Red
	C	Output signal from pickoff, second harmonic	Blue
	D	Output signal from pickoff, second harmonic	White
	E	Input power to motor, 8 Hz 34 ± 4V rms 2 phase	White/Gray
	F	Input power to motor, 8 Hz 34 ± 4V rms 2 phase	White/Black
	G	Internal ground	

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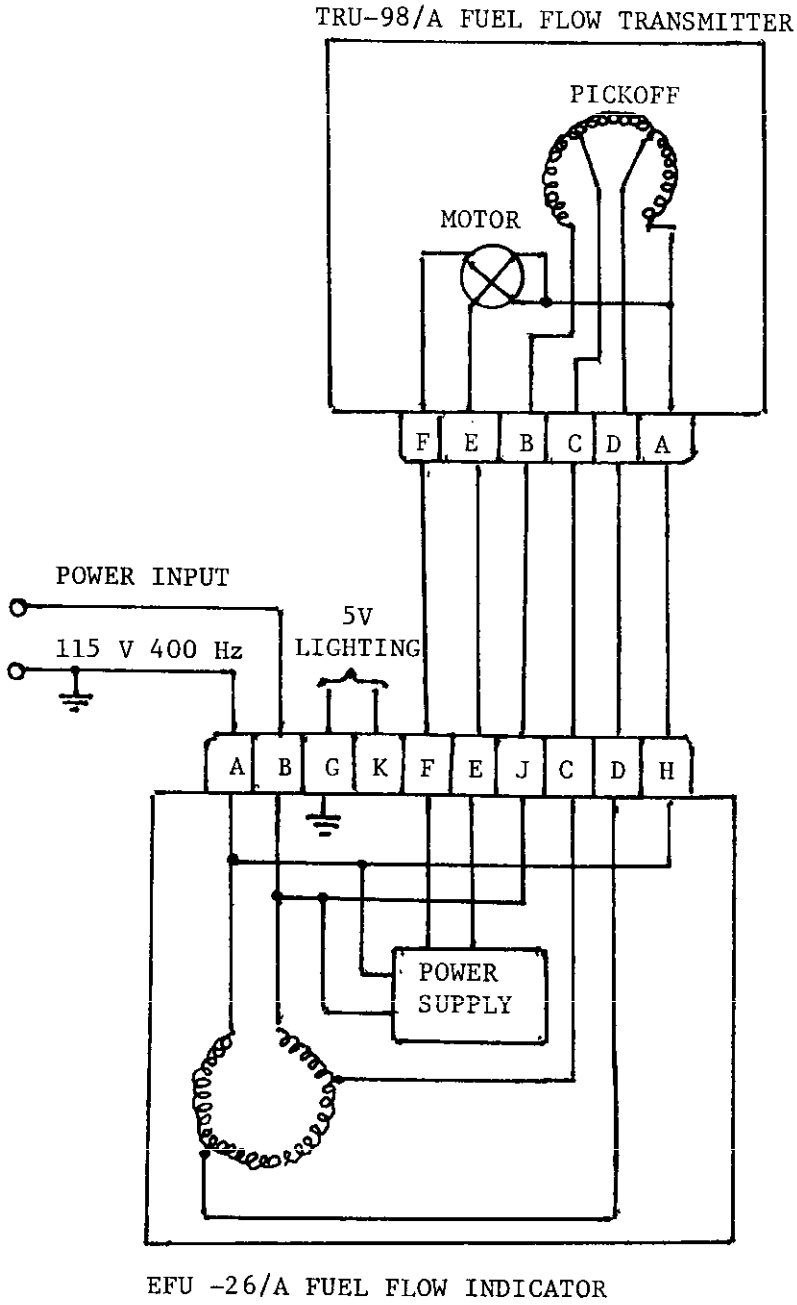
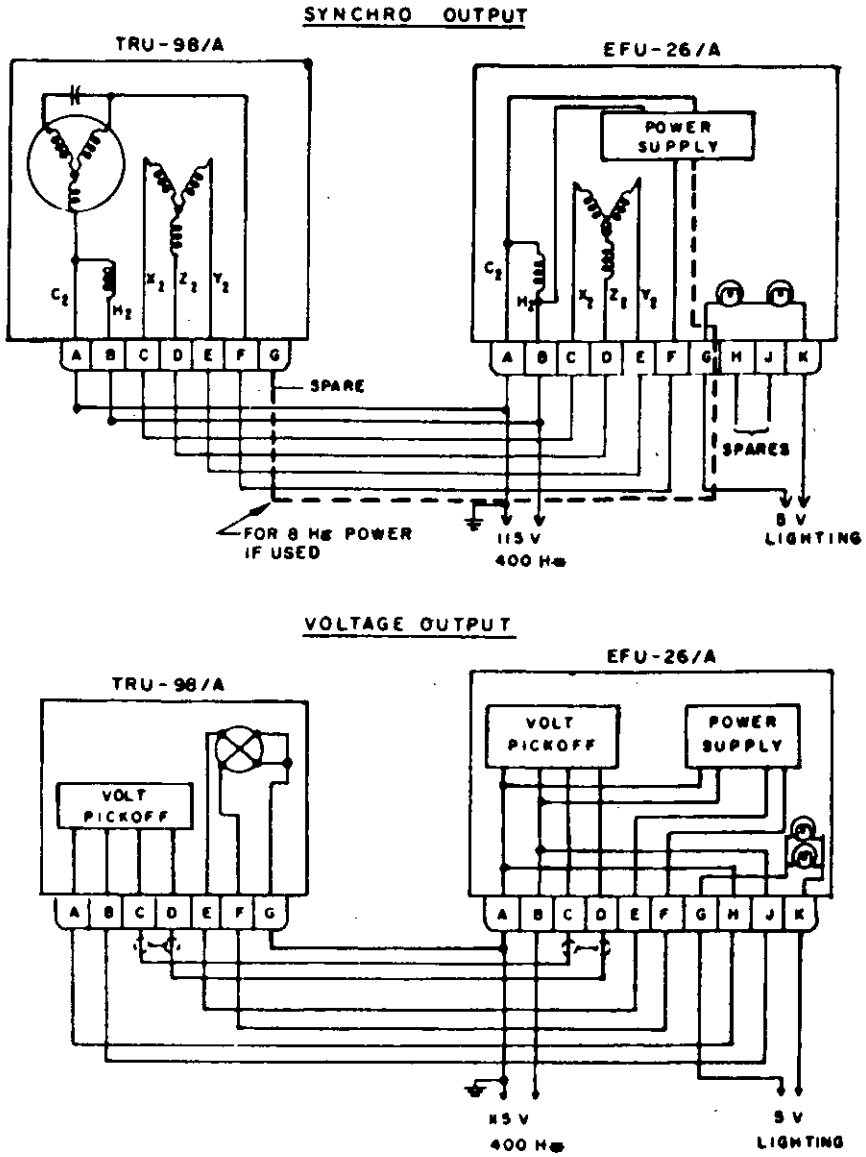


Figure 1. System connection diagram for TRU-98/A-4 and /A-7 transmitters.

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NOTE RECEPTACLE FOR TRU-98/A
CANNON GS02-16S-1P MATING CONNECTOR MS 3106E-16S-1P
RECEPTACLE FOR EFU-26/A
MS 3113-H12-10P MATING CONNECTOR MS 3126(E)12-10S
APPLIES TO SYNCHRO OUTPUT AND VOLTAGE OUTPUT

Figure 2. System connection diagram for TRU-98/A-1, /A-2, /A-3, /A-5 and /A-6 transmitters.

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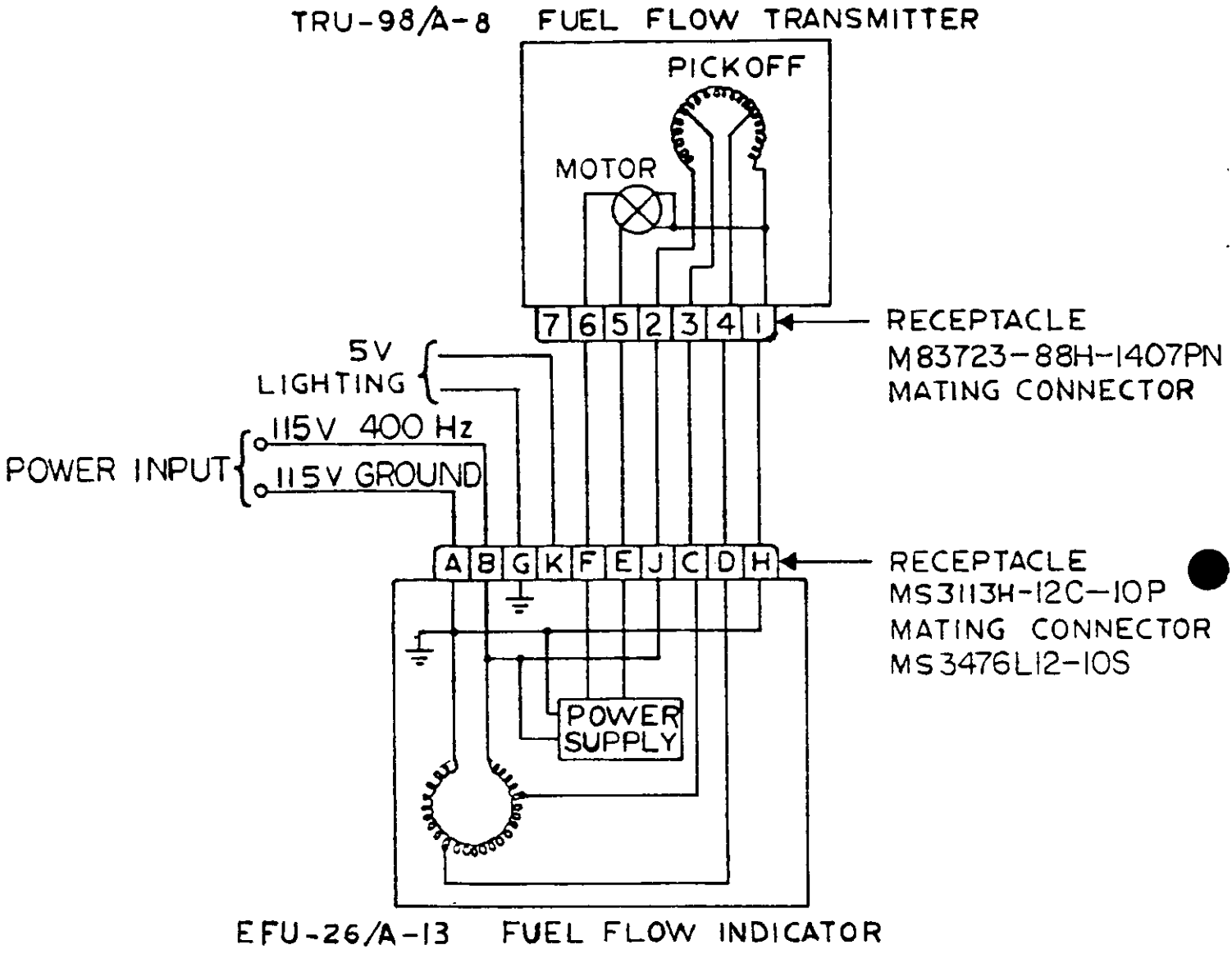


Figure 3. System Connection Diagram for TRU-98/A-8 Transmitter.

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3.5 Performance. The transmitter shall be capable of meeting the requirements specified herein under the following conditions:

a. Temperatures - operating temperatures ranging from -54° to $+121^{\circ}$C and storage temperatures ranging from -62° to $+150^{\circ}$C. Fuel ambients through the transmitter shall range from -54° to $+71^{\circ}$C.

b. Vibration - vibration from 0.1 double amplitude to $\pm$20g at frequencies from 5 to 2,000 Hz.

c. Altitude - altitudes ranging from sea level to 80,000 feet.

d. Shock - shock forces up to 15g each applies within a 7 $\pm$ 2 milliseconds time period.

e. Salt fog - salt fog for a period of 50 hours.

f. Humidity - relative humidity up to 95 percent.

g. Fungus - fungus growth as encountered in tropical climates.

h. Voltage and frequency variation - voltage and frequency variation ranging from 104V 360 Hz (320 Hz for the TRU-98/A-7 transmitter) to 126V 440 Hz (480 Hz for the TRU-98/A-7 transmitter).

i. Dielectric strength - application of 500V at a commercial frequency between any pin on the electrical receptacle and the transmitter case for 5 seconds.

j. Voltage surges - up to 195V.

k. Frequency cycling - from 320 to 480 Hz

3.5.1 Flame. The TRU-98/A-1 through /A-6 transmitters shall withstand a $1,093^{\circ}$C flame for 5 minutes with a fuel flow of 6000 pph.

3.5.2 Electromagnetic interference. The transmitter shall meet the electromagnetic interference requirements of MIL-STD-461 for class A1 equipment, except the frequency range shall be from 0.15 to 150 MHz.

3.5.3 Tolerances. When subjected to the tests specified herein, the flow rates for the transmitter specified shall be within the tolerances shown in the applicable columns of tables I, III, and IV.

* 3.5.4 Response. For the TRU-98/A-1 through /A-6 and /A-8 transmitters, the time required for the transmitter output signal to come within 1 percent of the final steady state shall not exceed 4 seconds under the conditions specified in 4.6.8. The TRU-98/A-7 transmitter response time shall not exceed 2 seconds under the same conditions. The TRU-98/A-7 transmitter output signal, when operating into an appropriate signal conditioner, shall be suitable for operating with an X-Y plotter. System response as displayed on the X-Y plotter, with calibrated overlays, shall be capable of following a ramp rate of flow having a slope of 8,500 pph per second not to exceed 3 percent over the basic transmitter tolerance at any point on the ramp above 1,500 pph. The ramp input can be initiated at any point above 300 pph.

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*TABLE III. Transmitter fuel flow rate tolerance in PPH

TRU-98/A-1 thru/A-6 fuel flow test points	8,000 PPH Range		10,000 PPH Range		12,000 PPH Range		15,000 PPH Range		20,000 PPH Range		25,000 PPH Range		TRU-98/A-7 and /A-8 fuel flow test points	15,000 PPH Range	
PPH	RM	L&H	RM	L&H	RM	L&H	RM	L&H	RM	L&M	RM	L&M	PPH	RM	L&H
100	40	60											200	85	127
200	33	49	40	60	50	75	65	98	100	150	120	180	500 (3)	75	112
500 (2)	27	40	35	53	44	66	58	87	80	120	100	150	1,000	60	90
1,000	24	36	30	45	38	57	50	75	70	105	90	135	1,500 (3)	45	67
1,500 (2)	26	39	30	45	37	55	47	70	65	98			2,000	40	60
2,000	28	42	30	45	36	54	45	67	60	90	80	120	2,500 (3)	30	45
2,500 (2)	33	49	35	53	39	58	45	67					*3,350 (1,3)	30	45
3,000	38	57	40	60	42	63	45	67	60	90	70	105	4,000 (3)	50	75
4,000 (2)	70	105	50	75	52	78	55	82	60	90	70	105	5,000 (3)	155	232
5,000	95	142	100	150	62	93	65	98	70	105	70	105	6,000	165	247
6,000 (2)	115	172	110	165	80	120	80	120	80	120	90	135	8,000 (3)	190	285
7,000 (2)	125	187	115	172	134	201	155	233					10,000	215	322
8,000	135	202	120	180	136	204	160	240	100	150	100	150	12,000 (3)	235	352
10,000 (2)			160	240	166	250	175	270	180	270	125	190	14,000	260	390
12,000 (2)					180	270	190	285	210	315	150	230	15,000 (3)	275	412
14,000							220	330	240	360	270	400			
15,000 (2)							225	340							
16,000									260	390	300	450			
18,000									300	450	339	500			
20,000									310	460	350	520			
25,000											400	600			

(1) *3,350 PPH test point corresponds to standard day (TRU-98/A-7)
(2) Denotes test point for TRU-98/A-4 individual room temperature scale error test
(3) Denotes test point for TRU-98/A-7 individual room temperature scale error test

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TABLE IV. Test tolerances in ± PPH.

TEST	TRU-98/A-1 Range		TRU-98/A-2 Range		TRU-98/A-3 Range		TRU-98/A-4 Range		TRU-98/A-5 Range		TRU-98/A-6 Range		TRU-98/A-7 & A-8 Range	
	High	Low	High	Low	High	Low	High	Low	High	Low	High	Low	High	Low
Voltage variation and frequency	32	8	40	10	48	12	60	18	80	20	100	25	60	10
Position	50	15	50	55	16	60	18	18	70	20	90	28	60	15
Friction														
Room and high temp	30	10	50	10	55	12	100	18	70	20	90	28	90	10
Low temp	50	15	50	15	55	16	100	18	70	25	90	30	90	15
Vibration														
Signal oscillation	140	40	140	40	160	45	160	50	180	55	200	60	170	40
Signal variation	140	40	140	40	160	45	160	50	180	55	200	60	170	40

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3.5.5 Power consumption. The power consumption of the transmitter motor shall not exceed 20.0 VA with rated voltage and frequency supplied and the power consumption of the transmitter pickoff shall not exceed 17 VA with rated voltage and frequency supplied.

* 3.5.6 Ranges. The transmitters shall measure mass flow rates in the following ranges:

a. TRU-98/A-1: From 0 to 3,000 pph and from 3,000 to 8,000 pph.

b. TRU-98/A-2: From 0 to 4,000 pph and from 4,000 to 10,000 pph.

c. TRU-98/A-3: From 0 to 6,000 pph and from 6,000 to 12,000 pph.

d. TRU-98/A-4: From 0 to 6,000 pph and from 6,000 to 15,000 pph.

e. TRU-98/A-5: From 0 to 8,000 pph and from 8,000 to 20,000 pph.

f. TRU-98/A-6: From 0 to 12,000 pph and from 12,000 to 25,000 pph.

g. TRU-98/A-7: From 0 to 4,000 pph and from 4,000 to 15,000 pph.

h. TRU-98/A-8: From 0 to 4,000 pph and from 4,000 to 15,000 pph.

3.5.7 Contaminated fuel. The transmitter shall be capable of operating one hour at 2000 pph as specified herein with contaminated fuel containing foreign particles of the sizes shown in table V.

3.5.8 Friction error. Friction error shall be within the tolerances specified in table IV.

3.5.9 Repeatability. The transmitter output signal shall be repeatable within ± 35 pph of the final calibration value at the 200 pph test point for room temperature conditions. The total absolute error shall not exceed the tolerances specified in table III.

3.5.10. Icing. The TRU-98/A-7 transmitter indication shall not bind or stick under the conditions specified in 4.6.14.

TABLE V. Contaminated fuel particle size.

Foreign particle size (Microns)	Precent of total weight
0-5	39 ± 2
5-10	18 ± 3
10-20	16 ± 3
20-40	18 ± 3
over 40	9 ± 3
Through a 200 mesh screen	100 percent

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3.6 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of DoD-STD-100 shall govern the manufacturer's part numbers and changes thereto.

* 3.7 Weight. The weight of the TRU-98/A-1 through /A-8 transmitters shall not exceed 5.0 pounds.

3.8 Mounting. For the TRU-98/A-1 through /A-6 transmitters, the transmitter end fittings having O-ring seals shall be designed for connection to the threaded fittings of the flared or flareless type (1-1/4 inch tubing size) having O-ring seals. The transmitter boss shall be threaded in accordance with MS33649-20. Mounting requirements for the TRU-98/A-7 transmitter shall conform to figure 4.

* 3.9 Dimensions. The outline dimensions of the TRU-98/A-7 transmitter shall conform to figure 4 and the outline dimensions of the TRU-98/A-1 through /A-6 transmitters shall conform to figure 5. The outline dimensions of the TRU-98/A-8 transmitter shall conform to figure 6.

3.10 Synthetic jewel bearings. When synthetic sapphire or ruby jewel bearings are used, they shall be in accordance with MIL-B-27497.

3.11 Flow direction. The direction of flow through the transmitter shall be permanently and legibly marked by an arrow on both the front and back of the transmitter.

3.12 Caps. The fuel-line connections shall be provided with protective shipping caps or plugs to prevent foreign material from entering the transmitter.

3.13 AGE compatibility. The prospective supplier shall submit a plan to the procuring activity which shall verify that the TRU-98/A-7 transmitter indication accuracy, when the transmitter is connected to the test equipment readout, shall not exceed the basic room-temperature-tolerance accuracies specified in table III by 2.0 pph in the low range nor 12.0 pph in the high range. The system readout accuracy shall be computed on the basic of root-sum-square (rss) of all the error sources including the transmitter and readout.

3.14 Finishes and protective coatings.

3.14.1 Alloy parts. Where practicable, aluminum or titanium alloy parts shall be covered with an anodic film conforming to MIL-A-8625. Small holes, inserts, and small areas that must be machined after finishing need not be covered with a protective film.

3.14.2 Chemical treatment. Where the primary purpose of the treatment is to afford a suitable paint base, or for castings containing nonaluminum or titanium alloy inserts, a chemical treatment in accordance with MIL-C-5541 may be used in lieu of an anodic film for alloys that do not anodize satisfactorily.

3.14.3 Abrasion resistance. Where abrasion resistance is a factor, chemical film in accordance with MIL-C-5541 shall not be used.

3.14.4 Magnesium parts. Magnesium alloy parts shall be covered with an anodic film in accordance with MIL-M-45202, HAE process.

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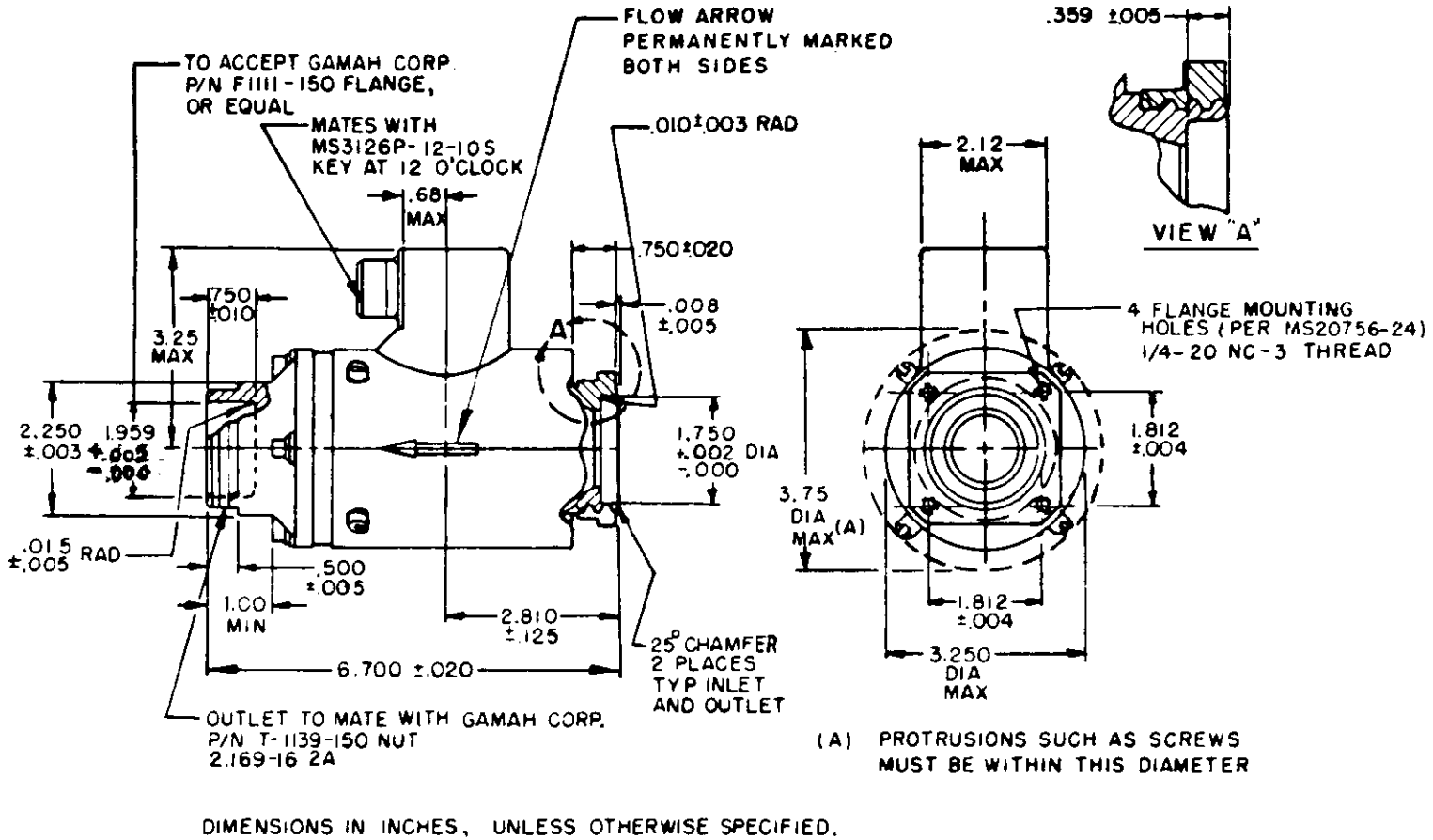


Figure 4. Outline dimensions and mounting provisions for the TRU-98/A-7 transmitter.

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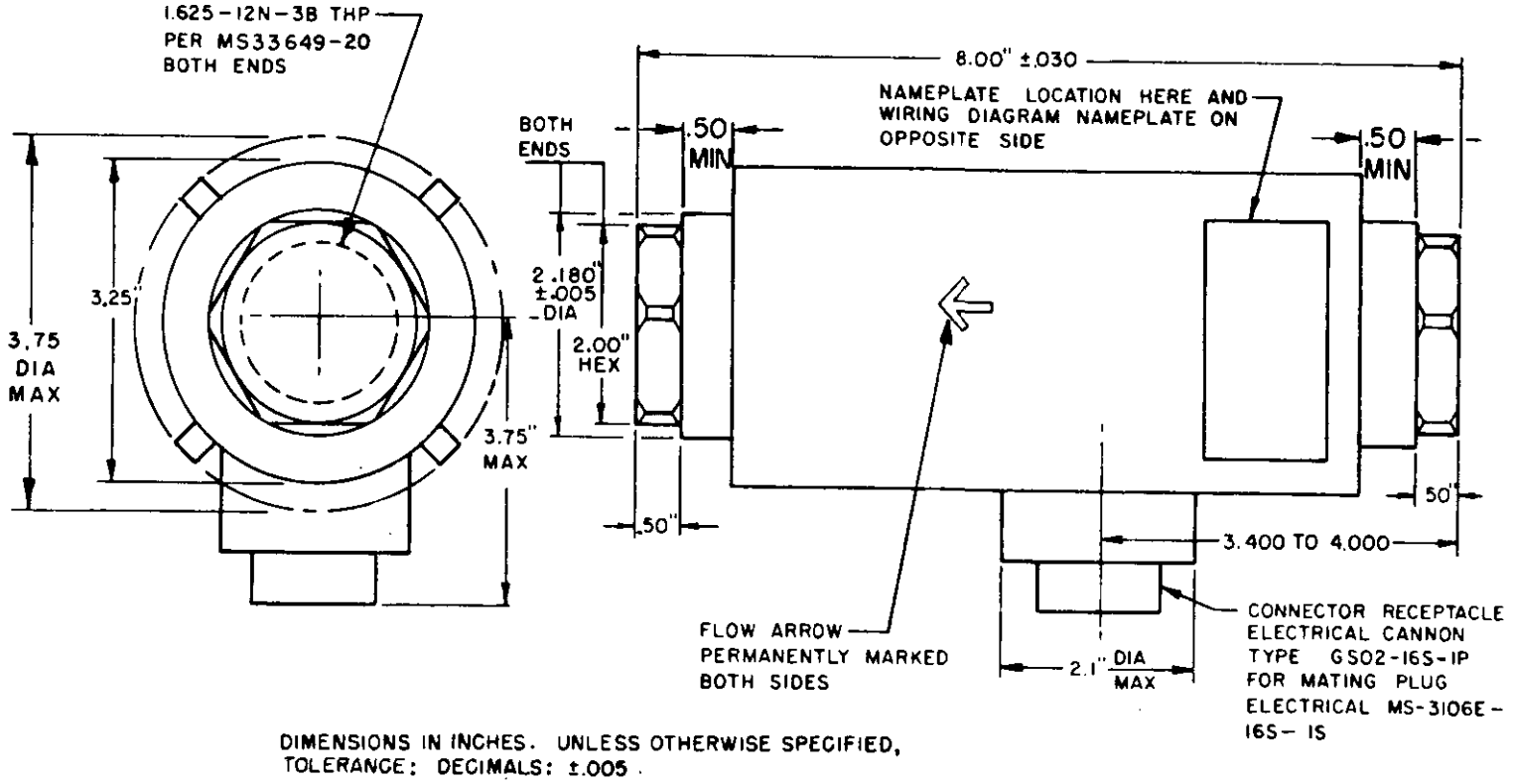


Figure 5. Outline dimensions and mounting provisions for the TRU-98/A-1 through /A-6 transmitters.

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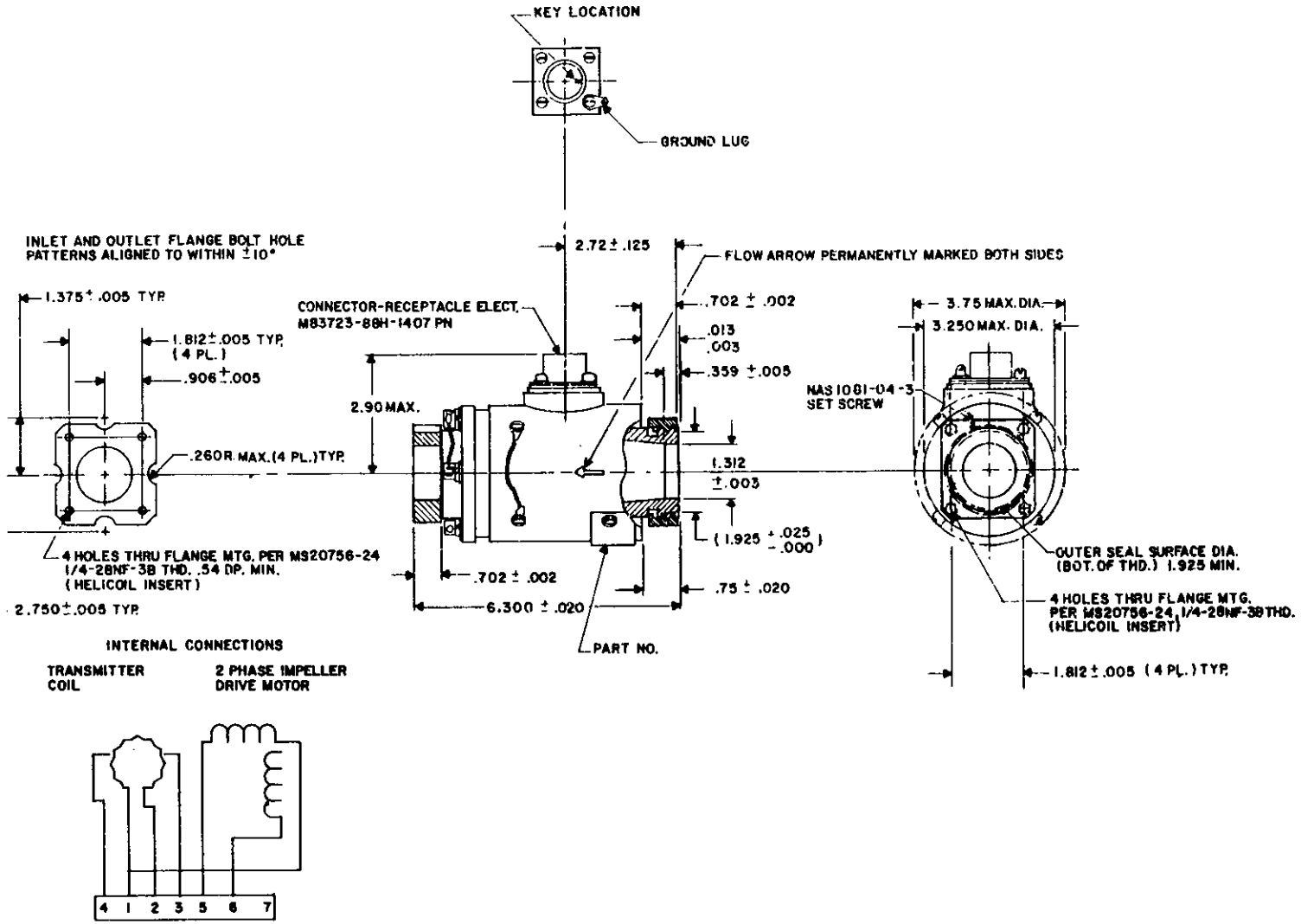


Figure 6. Outline dimensions and mounting provisions for TRU-98/A-8.

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3.15 Identification of product. Euqipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.15.1 Nameplate. A nameplate shall be provided in accordance with MIL-STD-130. Metal decalcomanias shall not be used.

3.15.2 Schematic. An electric schematic showing the function of each connector pin shall be permanently attached to the body of the transmitter.

3.16 Workmanship. The transmitter shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, impregnation of coils, welding, riveting, brazing, plating, painting, machine screw assemblies, and freedom of parts from burrs and sharp edges. Gears shall be of high quality and shall provide a minimum of backlash necessary for smooth operation. Where necessary, antibacklash techniques shall be used.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of tests. The inspection and testing of transmitters shall be classified as follows:

a. First article tests

b. Quality conformance tests

4.3 Test conditions.

4.3.1 Test power. The supply power shall be single phase 115V ± 2 percent, and 400 Hz controlled to less than ±1 Hz and of known value.

4.3.2 Test fuel. The fuel used for calibration in testing shall conform to MIL-C-7024C Type II.

4.3.3 Test position. Transmitters TRU-98/A through /A-6 shall be tested with the flow axis horizontal and the connector at the 3 o'clock position when viewed from the inlet end. The test position for the TRU-98/A-7 and /A-8 transmitters shall be with the connector at the 12 o'clock position when viewed from the inlet end.

4.3.4 Standard atmospheric conditions. Whenever the pressure and temperature existing at the time of the tests are not specified definitely, it is understood that the test is to be made at atmospheric pressure (approximately 29.92 inches Hg) and room temperature (approximately 25°C). When tests are made with

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ataospheric pressure or room temperature differing materially from the above values, proper allowances shall be made for the difference from the specified conditions.

4.3.5 Vibration. Unless otherwise specified, the transmitter shall be vibrated before a test reading is taken. An electro-mechanical or pneumatic vibrator, approximating normal engine vibration and attached to the flow pipe, may be used.

4.3.6 Test equipment. Unless otherwise specified, the transmitter shall be tested while connected to a master indicator and master power supply. The master power supply shall be such that it will have no frequency error, or if a frequency error should exist, it shall be subtracted from the test data. All flow tests shall be performed on a primary flow stand which measures liquid mass flow rate directly in terms of weight and time, or by a means directly related (no more than once removed) to the weight and time standard.

* 4.3.6.1 Synchro output (transmitter output indications). A phase sensitive nullmeter with damping and related null balance equipment shall be standard readout equipment for the fuel flow transmitter. All first article and sampling plan B tests shall be performed on a primary flow stand which measures liquid mass flow rate directly in terms of weight and time. The combined test equipment shall have a total maximum error of 0.10 circular degree.

4.3.6.2 Voltage output.

4.3.6.2.1 Master voltmeter. The output signal from the transmitter shall be measured with a master voltmeter especially designed for the purpose and accurate to within $\pm$0.1 percent. The input impedance of the voltmeter shall be 100,000 ohms minimum resistive load. The voltmeter shall be calibrated in rms voltage. The voltmeter reading shall be proportional to the peak value of the AC voltage. Calibration of the voltmeter shall be established by comparison with an electro-dynamometer type voltmeter calibrated with a pure sine wave. As an alternate, the master voltmeter may consist of a transducer nulled against the output signal.

4.3.6.2.2 Phase angle meter. The phase angle meter shall be especially designed for the purpose and shall be accurate to $\pm 0.1^{\circ}$.

4.3.7 Tolerances. Test tolerances shall be as specified in table I, III, and IV.

* 4.4 First article testing.

4.4.1 Test samples. The test samples shall consist of at least ten transmitters representative of the production equipment. Three of the samples shall be subjected to the sampling plan A and B tests and seven samples shall be subjected to the sampling plan C tests. The samples shall be identified with the manufacturer's part number.

4.4.2 Test report. The contractor shall prepare a test report in accordance with MIL-STD-831 (see 6.2.1).

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* 4.4.3 First article tests. The first article tests shall consist of all the tests specified under 4.6. Unless otherwise specified by the procuring activity, tests shall be conducted in the order listed.

4.5 Quality conformance tests. Quality conformance tests shall consist of:

a. Individual tests

b. Sampling tests

4.5.1 Individual tests. Each transmitter shall be subjected to the following tests as described under 4.6:

a. Burn-in

b. Leakage

c. Examination of product

d. Dielectric strength

e. Scale error

f. Friction error

g. Position error

4.5.2 Sampling tests.

4.5.2.1 Sampling plan A. A minimum of 3 transmitters shall be selected at random from each 100 or fraction thereof produced and subjected to the following tests as described under 4.6:

a. Individual tests

b. Voltage and frequency variation

c. Voltage surge

d. Power consumption

e. Low temperature

f. High temperature

g. Vibration error

h. Pressure drop

4.5.2.2 Sampling plan B. Unless otherwise specified (see 6.2), 3 transmitters shall be selected at random from the first 10 produced on the contract or order and subjected to the following tests as described under 4.6. If any one of the samples specified in sampling plan A is selected for samping plan B, the sampling plan A tests shall not be repeated on that sample.

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a. Sampling plan A tests

b. Low temperature exposure

c. Temperature - altitude

d. High temperature exposure

e. Altitude

f. Pressure leakage - high temperature

g. Humidity

h. Fungus

i. Salt fog

j. Electromagnetic interference

k. Shock

l. Contaminated fuel

m. Substitute fuel

n. Frequency cycling

o. Flame

p. Vibration endurance

q. Icing

4.5.2.2.1 Test schedule. When sampling plan B tests are not specifically waived on a contract, the tests shall begin as soon as the first ten items have been produced but before any production deliveries have been made (see 6.2.1).

4.5.2.3 Rejection and retest. When an item selected from a production run fails to meet the specification, no items still on hand or later produced shall be accepted until the extent and cause of failure have been determined and appropriately corrected. The contractor shall explain to the Government representative the cause of failure and the action taken to preclude recurrence. After correction, such tests as determined by the procuring activity shall be repeated.

4.5.2.3.1 Individual tests may continue. For production reasons, individual tests may continue pending the investigation of a sampling test failure. But final acceptance of items on hand or produced later shall not be made until it is determined that all items meet all requirements of the specification.

4.5.2.4 Sampling plan C. Seven transmitters shall be selected at random from the first 14 produced and subjected to the following tests as described under 4.6. Rejection for reliability shall be in accordance with 4.6.33.6.

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a. Individual tests

b. Reliability

c. Longevity

4.5.2.4.1 Reliability test program. Within 90 days after receipt of contract or purchase order and 30 days prior to the scheduled date of initiation of testing, or at such later date as the contracting officer may authorize, the contractor shall submit a detailed reliability test program to the procuring activity for approval. Testing shall begin only after formal approval of this test program by the procuring activity.

4.5.3 Defects in item already accepted. The investigation of a test failure could indicate that defects may exist in items already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and the method of correcting them.

4.6 Test methods.

4.6.1 Examination of product. The transmitter shall be inspected to determine compliance with the requirements specified herein with respect to materials, finish, workmanship, and marking.

4.6.2 Dielectric strength. A potential of 500V at a commercial frequency shall be applied between any pin and the transmitter case for a minimum of 5 seconds. There shall be no insulation breakdown as a result of this test.

4.6.3 Scale error. The scale error of each transmitter shall be determined at the test points specified in table III at room temperature. The scale error shall be determined for both increasing and decreasing flow rates. Each test point in a set of readings shall be consistently approached from the same direction. A minimum of two sets of readings shall be taken at each test point. A set of readings shall consist of two 30 second flowstand weight cycle times (or equivalent, see 4.3.6) averaged to obtain average flow reates for two independent runs. The values for both runs shall be reported. If the individual readings for one of the runs is within 5 percent of the specification limit, a second reading shall be taken. Scale error shall be determined while the transmitter is gently vibrated by an electro-mechanical or pneumatic vibrator mounted on the flow piping. Scale error tests conducted at room, low, and high temperatures, and substitute fuel operating conditions shall be in accordance with this procedure. Room temperature scale error acceptance test for TRU-98/A-4 and the TRU-98/A-7 transmitters shall be performed at flow points identified in table III. Test tolerances for all scale error tests shall be in accordance with table III.

* 4.6.4 Friction error. Friction error shall be determined at 25, 50, and 90 percent of full scale for the TRU-98/A-1, /A-2, /A-4, /A-5, /A-6, and /A-7 transmitters, and 25 and 90 percent of full scale for the TRU-98/A-3 and /A-8 transmitters. Friction error testing on compliance with individual tests (see 4.5.1) shall be performed at the 4,000 and 14,000 flow points for the TRU-98/A-4 transmitter and at the 3,350 and 14,000 flow points for the TRU-98/A-7 and /A-8 transmitters.

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* 4.6.5 Leakage. The TRU-98/A-1 through /A-6 transmitters shall be tested for leakage as follows: A nonshock hydraulic pressure shall be applied to the inlet connection with the outlet connection bled free of air and capped. A pressure of 1,600 psi shall then be rapidly applied five times at room temperature. After the last application of pressure and while still at the test pressure and temperature, there shall be no indication of leakage for a period of 5 minutes as checked on a pressure gage readable to 10 psi or by visual leakage of a colored test fluid. The transmitter temperature shall then be reduced to $-54^{\circ} \pm 2^{\circ}C$ by submersion in a cold liquid bath, already at this temperature, for a period of not less than 15 minutes. While still at this temperature, the test pressure shall be applied rapidly five times. After the last application of pressure, there shall be no evidence of leakage for a period of 5 minutes as detected by the same means used for the room temperature portion of this test. The TRU-98/A-7 and /A-8 transmitters shall also be subjected to this test, except the pressures shall be 120 psi for the /A-7, and 1360 psi for the /A-8 transmitter.

4.6.6 Burn-in. The transmitter shall be placed in a chamber with the temperature maintained at $50^{\circ}C$ for a minimum of 24 hours with fuel flowing equivalent to a half-scale indication. At least 8 hours of this test shall be performed during normal working hours. Observations shall be made and recorded at least once every 2 hours throughout the 8-hour period. At no time shall the transmitter output change by more than one-half the tolerance specified for the test point in the applicable tolerance table. Pointer oscillation or erratic movement shall be cause for rejection. The transmitter shall be returned to room temperature for a minimum of 2 hours after which it shall meet the test specified in 4.6.3. Burn-in testing for acceptance shall be conducted at room temperature with the transmitter filled and capped with clean fuel. The motor and pickoff coil shall be energized and operated for a period of at least 24 hours. The output signal of the start and at the end of the burn-in test shall be recorded along with the total number of burn-in hours. The transmitter output shall not change by more than 30 pph. The transmitter shall be rejected if this is exceeded.

4.6.7 Position error. Position error shall be determined at 25 to 90 percent of full scale. Readings shall be taken in the normal position and with the transmitter turned 90° in any direction. The reading taken in any position shall not differ from the reading taken in the normal position by more than the value specified in table IV.

4.6.8 Response time. The time required for the transmitter output signal to come within 1 percent of the final steady state signal shall not exceed 4.0 seconds (2.0 seconds for the TRU-98/A-7) when a step change in flow rate is applied. The flow rate shall be changed from approximately 2 percent of full scale to a full scale indication by activation of a quick opening valve. The timer will be started simultaneously with the activation of the valve.

4.6.9 Voltage and frequency variation. The applied voltage and frequency for this test shall be set at the following points:

	Voltage	Frequency
(Normal Condition)	115	400
	104.0	320
	104.0	480
	126.0	320
	126.0	480

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The voltage and frequency test shall be conducted at 25 and 90 percent of full scale flow rates. Readings shall be taken at the 115V, 400 Hz normal condition and at the conditions specified herein. The reading taken at each of the above conditions shall not differ from the reading taken at the normal condition in excess of the value shown in table IV. The transmitter shall be operated at least 1/2 hour at each of the specified conditions before any readings are taken. These tests shall be run with a master indicator and a master power supply (see 4.3.6).

4.6.10 Pressure drop. The pressure drop across the transmitter shall be determined at the flow rates specified in table I. The pressure drop shall be recorded both with the transmitter energized and unenergized. The pressure drop shall be obtained by sensing the pressure at the inlet and outlet ends of the flowmeter and then obtaining the difference between the two readings. The pressure drop shall not exceed the values specified in table I.

4.6.11 Voltage surge. A master indicator need not be used during the surge application of this test. A second harmonic receiver, or simulated receiver, compatible with the transmitter will be adequate. A master power supply must be used. The tests shall be made at approximately 50 percent of full scale. Power shall be applied at 115 ± 1V, 400 ± 2 Hz for 1 hour. At the end of this operating period, the power input shall be adjusted to 115 ± IV, 320 ± 2 Hz. A voltage surge shall then be applied to provide an rms peak voltage of 160 ± 5V and return to 115 ± IV. The duration of the surge including rise and decay time shall be 0.09 ± .005 second. The power input shall then be adjusted to 115 ± IV, 480 ± 2 Hz and a voltage surge applied to provide an rms peak voltage of 195 ± 5V and return to 115 ± IV. The duration of this surge including rise and decay time shall be 0.07 ± .005 second. At the conclusion of this procedure, a scale error test at room temperature shall be conducted. The errors shall not exceed the room temperature tolerances of table III.

4.6.12 Vibration error. The vibration error test shall be conducted in accordance with MIL-STD-810, method 514, curve AT, at room temperature with fuel flowing at approximately 3,000 and 10,000 pph. For sampling plan A only, the vibration shall be limited to one sweep cycle of 5 to 2,000 to 5 Hz in 15 minutes for each flow rate. The errors due to vibration shall not exceed those shown in table IV.

4.6.13 Vibration endurance. The transmitter shall be subjected to an endurance cycling test for 40 hours in each of the three mutually perpendicular axes. The duration of one cycle (5 to 2,000 to 5 Hz) shall be approximately 30 minutes. The vibration spectrum shall be in accordance with MIL-STD-810, method 514, equipment category (b), curve L. After completion of this test, a scale error test shall be conducted. The errors shall not exceed the high and low temperature limits specified in table III. The transmitter shall be completely disassembled and examined for damage or wear that might result in a failure when projected over the design life of the instrument.

4.6.14 Icing. Transmitters that are to be airframe mounted shall be installed in suitable test equipment at room temperature. Fuel with a water concentration of 0.054 percent by volume shall be reduced to a temperature of -5° to -10° C and passed through the transmitter at an indicated flow rate of approximately 8,000 pph for a period of 5 minutes. The indicated flow rate shall not change by more than ± 100 pph. While still at the low temperature, the flow rate shall be increased to 15,000 pph and then reduced to 0. There shall be no evidence of binding or sticking of the indication.

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4.6.15 Power consumption. With 115V 400 Hz applied to the indicator and power from the indicator to the transmitter motor applied to the transmitter, power consumption of the transmitter motor shall not exceed 20 VA and power consumption of the transmitter pickoff shall not exceed 17 VA.

4.6.16 Low temperature. The transmitter shall be placed in a test chamber and maintained at a temperature of $-54^{\circ} \pm 2^{\circ}C$ for a period of 4 hours. At the end of this period and while still at the specified low temperature, the transmitter shall be subjected to the tests specified in 4.6.3 and 4.6.4. The errors shall not exceed the tolerances specified in tables III and IV, respectively.

4.6.17 High temperature. The transmitter, while inoperative, shall be subjected to a temperature of $121^{\circ} \pm 2^{\circ}C$ for a period of 4 hours after which it shall be tested at the specified high temperature for scale error and friction error as specified in 4.6.3 and 4.6.4. The errors shall not exceed the tolerances specified in tables III and IV, respectively.

4.6.18 Low temperature exposure. The transmitter shall be placed in a test chamber and the chamber cooled to and maintained at a temperature of $-62^{\circ}C$ for a period of 72 hours at which time the equipment shall be inspected for deterioration. The temperature of the chamber shall then be raised to $-54^{\circ}C$ and maintained for an additional 24-hour period. At the end of the 24-hour period and while still at $-54^{\circ}C$, the transmitter shall be subjected to the scale error and friction error tests. The transmitter shall then be maintained at room temperature for a period of not less than 4 hours after which it shall be subjected to the scale error and friction error tests at room temperature. The transmitter shall be examined and there shall be no deterioration of nor damage to the transmitter as a result of the extreme temperature exposure.

4.6.19 Temperature altitude. The transmitter shall be placed in a test chamber and the chamber cooled to and maintained at a temperature of $-54^{\circ} \pm 2^{\circ}C$ and a pressure equivalent to a 65,000-foot altitude for a period of 4 hours. While at this temperature and pressure, the transmitter shall be tested for scale error at 0 pph. This test may be combined with the test specified in 4.6.16. There shall be no damage to nor deterioration of the transmitter as a result of this test.

4.6.20 High temperature exposure. While inoperative the transmitter shall be subjected to a temperature of $121^{\circ} \pm 2^{\circ}C$ for a period of 4 hours. The temperature shall be raised to $150^{\circ} \pm 2^{\circ}C$ for 30 minutes. This cycle shall be repeated 6 times for a total of 24 hours at $121^{\circ} \pm 2^{\circ}C$ and 3 hours at $150^{\circ} \pm 2^{\circ}C$. The transmitter shall then be energized and immediately tested for scale error and friction error at $121^{\circ} \pm 2^{\circ}C$ ambient air and $38^{\circ} \pm 2^{\circ}C$ fuel. The tolerances shall not exceed those specified in tables III and IV. The transmitter shall be examined and there shall be no evidence of failure. During temperature cycling, fluid may be inside the transmitter but not flowing.

4.6.21 Altitude. The transmitter shall be filled with fuel, placed in an altitude chamber, and subjected to a pressure equivalent to an altitude of 80,000 feet for a period of 4 hours. During this period, the transmitter shall be energized and tested for room temperature scale error at 0 pph. Scale error shall be within the tolerance specified in table III.

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*4.6.22 Pressure leakage-high temperature. The transmitter shall be subjected to a pressure leakage at high temperature test as follows: A nonshock pressure shall be applied to the inlet connection with the outlet connection bled free of air and capped. The transmitter shall be heated in a 140°C chamber for 1/2 hour and a pressure of 1,000 psi (-12 and +200 psi for the TRU-98/A-7 transmitter; 800 psi -12 and +200 psi for the /A -8 transmitter) shall then be applied rapidly five times. After the last application of pressure and while still at the test pressure and temperature, there shall be no indication of leakage for 5 minutes as checked by a pressure gage readable to 10 psi or by visual leakage of a colored test fluid.

4.6.23 Humidity. The transmitter shall be subjected to humidity test in accordance with MIL-STD-810, method 507, procedure I. After completion of the cycling, the transmitter shall meet the tests specified in 4.6.3 and 4.6.4.

4.6.24 Fungus. The transmitter shall be subjected to a fungus test in accordance with MIL-STD-810, method 508, procedure I. At the conclusion of this test, the transmitter shall be visually examined. There shall be no deterioration or corrosion of internal components as a result of this test.

4.6.25 Salt fog. The transmitter shall be subjected to a salt fog test in accordance with MIL-STD-810, method 509, procedure I, for a period of 50 hours. The scale error and friction tests shall be conducted before and after this test.

4.6.26 Electromagnetic interference. The electromagnetic test shall be conducted in accordance with MIL-STD-462 except the frequency range shall be from 0.15 to 150 MHz. Test methods shall be as follows: CE02, CE03, CE04, CSOI, CS02, CS06, RSOI, RS02, and RS03.

4.6.27 Shock. The shock test shall be in accordance with MIL-STD-810, method 516, procedure I, except as specified herein. The transmitter shall be subjected to three impact shocks of 15g's. Each shock shall be applied for 7 $\pm$ 2 milliseconds in each of the following directions:

a. Along the longitudinal axis in each direction

b. At 45 to the longitudinal axis in one direction

4.6.28 Contaminated fuel operation. Test fuel containing dirt to the extent of 80 grams of foreign matter to 1,000 gallons of fuel shall be circulated through the transmitter at approximately 2,000 pph for 1 hour. Scale error tests shall then be conducted using the contaminated fuel. Operation shall be satisfactory and within the room temperature tolerances. The foreign matter shall consist of not less than 68 percent SiO_2 and shall have a particle size analysis as shown in table V.

4.6.29 Substitute fuel. The substitute fuel test shall be conducted using JP-6 fuel as specified in MIL-T-5624. This test shall be conducted under room operating temperature. The scale errors shall not exceed the tolerances specified in table III for high and low temperatures.

4.6.30 Frequency cycling. The master indicator need not be used for the cycling portion of this test. A second harmonic receiver, or simulated receiver, compatible with the transmitter will be adequate. A master power supply shall be

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used. The voltage shall be adjusted to 115 ± IV and the frequency cycled from 360 (320 for the /A-7) ± 2 Hz to 440 (480 for the /A-7) ± 2Hz in accordance with the following schedule with approximately half scale flow. This constitutes one cycle for a total time of 100 seconds.

a. Start cycle at 360 Hz (320 Hz for the /A-7) increase frequency to 440 Hz (480 Hz for the /A-7) in 38 seconds linearly with time.

b. Return frequency to 360 Hz (320 Hz for the /A-7) in 2 seconds or less.

c. Increase frequency to 440 Hz (480 Hz for the /A-7) in 58 seconds linearly with time.

d. Return frequency to 360 Hz (320 Hz for the /A-7) in 2 seconds or less.

This cycle shall be repeated continuously for 10 hours at the rate of 36 cycles per hour. Power shall then be removed for a maximum of 2 hours and reapplied for 1 hour at 120 ± 1V, 320 ± 2 Hz. Immediately following the 1 hour of operation at 120V, 320 Hz, the transmitter shall be frequency cycled for 1 additional hour on the schedule indicated above, except that the frequency range shall be increased from 320 to 480 Hz. The transmitter shall then be subjected to the scale error test at room temperature. The scale error shall not exceed the tolerances specified in table III for room temperature.

4.6.31 Compatibility. Compatibility tests as required to demonstrate compliance with 3.4.2, 3.4.3, 3.4.4, 3.4.4.1, 3.4.4.2, 3.4.4.3, 3.4.4.4, 3.13, 4.6.2, 4.6.3, 4.6.4, 4.6.8, and 4.6.9 shall be conducted in accordance with the test procedures or the applicable contract document for the transmitters specified herein.

4.6.32 Flame. While inoperative but with a fuel flow of 6,000 pph, the transmitter shall be subjected to a 1,093°C flame for 5 minutes. Any physical damage that would result in leakage sufficient to sustain a flame shall be cause for rejection. When sampling plan B and C tests are conducted, this test shall be run after the longevity test on one transmitter only. This test is intended for engine mounted transmitters only.

4.6.33 Reliability. The seven transmitters shall be subjected to a reliability test in accordance with MIL-STD-781, test level G, test plan I, which shall demonstrate a minimum acceptable MTBF of 1,200 hours to a 90 percent confidence level.

4.6.33.1 Test equipment. The test equipment and fixtures shall include provisions for transmitter operation, temperature cycling, and vibration.

4.6.33.2 Operation. During this test, fuel at room temperature shall be flowing through the transmitter at a rate from 5 to 90 ± 5 percent of full scale and return at a rate of 2 ± 1 cycles per minute.

4.6.33.3 Temperature cycling. Unless otherwise specified or amended by the procuring activity due to the results of the thermal analysis, the temperature of the chamber in which the transmitters are tested shall be cycled as follows:

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a. Starting at room temperature, decrease temperature to -54°C and stabilize for 1 hour with transmitters unenergized.

b. Turn transmitters on, raise temperature to 95°C, and operate transmitters for 3 hours.

c. Turn transmitters off and lower chamber temperature to -54°C for 1 hour.

d. Repeat steps b and c until reliability requirements have been met. This schedule provides six 4-hour test cycles for each 24-hour period. In each 24-hour period, the operational test time is 18 hours.

4.6.33.4 Vibration cycling. Throughout this test, the transmitters shall be vibrated for a period of 10 minutes during each hour of on-time at any nonresonant frequency between 20 and 60 Hz and at a peak acceleration of 2.2g ± 10 percent.

4.6.33.5 Performance check. At each 200-hour interval and at the end of this test, the transmitters shall be subjected to the scale error, friction, and response time tests. During each scale error test, if any two test points exceed the allowable tolerances for high and low temperatures and substitute fuel specified in tables III and IV, one failure shall be changed against this test.

4.6.33.6 Failure criteria. For an accept decision, each transmitter shall be operated at least 1,000 hours. If a transmitter fails, entries shall be made on the appropriate data logs and the failed equipment shall be removed from the test and repaired with minimum interruption of the transmitters still under test. The procuring activity shall be immediately notified of the failure and the plan for correcting the deficiency. A detailed analysis of the failure will be submitted as soon as it has been completed. No more than a total of 12 failures will be permitted during this test. Acceptance shall be based on the accept-reject criteria specified in MIL-STD-781, test plan I, after each transmitter has been operated for at least 1,000 hours.

4.6.33.7 Reporting. Reporting shall be submitted in accordance with MIL-STD-781. The reliability test log and failure summary and analysis shall be included in the final test report (see 4.4.2).

4.6.34 Longevity. Testing on two of the seven reliability test samples shall be continued through a longevity test in accordance with MIL-STD-781, test plan XXVIII, until 2,500 hours have accumulated on each of the two transmitters.

4.7 Inspection of preparation for delivery. Test methods for preservation and packaging shall be accomplished in accordance with section 4 of MIL-P-116 to insure compliance with section 5 of this specification. Packaging shall be accomplished as specified herein.

* 5. PACKAGING

* 5.1 Preservation-packaging. Preservation-packaging shall be level A or C as specified.

* 5.1.1 Level A.

* 5.1.1.1 Cleaning. Transmitter, rate of flow shall be cleaned in accordance with process C-1 of MIL-P-116.

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* 5.1.1.2 Drying. Transmitter, rate of flow shall be dried in accordance with the appropriate process of MIL-P-116.

* 5.1.1.3 Preservation application. Not applicable.

* 5.1.1.4 Unit packaging. Unless otherwise specified by the procuring activity, each transmitter, rate of flow shall be packaged in quantity unit packs of one each in accordance with Method IC-1 of MIL-P-116. The completed pack shall be overboxed in containers conforming to PPP-B-636. Apply sufficient cushioning material between bag and unit container of a type, density and thickness to insure shock transmission does not exceed peak values in G's established for the transmitters when completed packs are subjected to the rough handling drop tests of MIL-P-116.

* 5.1.2 Level C. Same as Level A except Method III of MIL-P-116 applies in lieu of specified method.

* 5.2 Packing. Packing shall be Level A, B or C as specified by the procuring activity.

* 5.2.1 Level A. Transmitter, rate of flow, packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, weather-resistant class, unless otherwise specified by the procuring activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, and minimum tare and cube consistent with the protection required. Table II of MIL-STD-794 applies for Army use.

* 5.2.2 Level B. Transmitter, rate of flow, packaged as specified in 5.1.1 shall be packed in domestic class exterior shipping containers conforming to PPP-B-636. Other requirements as specified in 5.2.1 above shall apply.

* 5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirements of this level.

5.3 Marking of shipments. Marking of shipments shall be in accordance with MIL-STD-129. The nomenclature shall be as follows:

Transmitter, Rate of Flow
TRU-98/A*____ **_____PPH

* Add dash number as applicable
** Add range as applicable

5.3.1 Reinspection date markings. The reinspection date markings shall be as specified by the procuring activity.

6. NOTES

6.1 Intended use. The TRU-98/A transmitters covered by this specification are intended for use in measuring and transmitting a signal proportional to the total mass rate of fuel flow to the main engine of jet aircraft.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Type of transmitter required (see 1.2).

c. When sampling plan B tests will not be conducted (see 4.5.2.2).

d. That the items specified in 4.5.3 be returned to the contractor for appropriate corrective action at the contractor's expense.

e. Levels of packaging and packing required and reinspection date markings (see section 5).

6.2.1 Data and hardware. The contractor will submit the following data and hardware to the procuring activity:

a. Outline drawing of the transmitter showing dimensions, display, et cetera, and all other configuration details required to demonstrate compliance with this specification.

b. Internal circuit schematic and interconnecting diagram.

c. Installation instruction sheet.

d. Three copies of the test report specified in 4.4.2, signed by a Government inspector, demonstrating compliance of the transmitter with all the tests specified herein.

e. All test samples in the "as is" condition.

f. Test box or wiring harness for laboratory testing.

g. Test data for sampling plan B tests to be submitted to the procuring activity no later than 16 weeks after starting the test program.

6.3 Test procedures. The first article test procedures will be submitted for formal approval before the test program is initiated.

6.4 Test samples. Unless otherwise specified, the first article test samples will be retained by the procuring activity.

6.5 Marginal indicia. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in those notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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* 6.6 First article. When a first article is required, the item will be inspected and tested in accordance with Section 4 of this document. The contracting officer should include specific instructions in acquision documents regarding arrangements for examinations, test and approval of the first article.

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Preparing activity
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1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
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	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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